ORF307
Multi-objective least squares

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Today's topics

- Multi-objective least squares
- Control
- Estimation
- Regularized data fitting

- **Multi-objective least squares**

Multi-objective least squares

- **Fact of life**: we have to consider more than one objectives $(k \geq 2)$

$$J_1 = ||A_1 x - b_1||^2, J_2 = ||A_2 x - b_2||^2, \ldots, J_k = ||A_k x - b_k||^2$$

  - $A_i \in R^{m_i \times n}$ and $b_i \in R^{m_i}$ for $i = 1,2, \ldots k$

- **Case #1**: we can minimize each objective individually, resulting in the solution of k different optimization problems and k different solutions

- **Case #2**: we can solve one optimization problem that makes all objectives as small as possible simultaneously, resulting in only one solution

Remark: In this case the optimal solution can be thought of as a compromise for all objectives

Weighted sum of the objectives

- Form a new objective that consists of the **weighted sum of the individual objectives**

$$J = \lambda_1 J_1 + \lambda_2 J_2 + \cdots + \lambda_k J_k$$

- Solve the **single objective** least squares problem to obtain x^*

$$\min J = \lambda_1 J_1 + \lambda_2 J_2 + \cdots + \lambda_k J_k$$

- The **weights** λ_i represent the **importance of each objective**
  - If $\boldsymbol{\lambda_i = 1, \forall i}$ then each objective is **equally important**
  - **Bi-objective** optimization problem: $\min J = J_1 + \lambda J_2$

  Remark: if $\lambda \approx 0$ we focus on minimizing J_1 ; if $\lambda \gg 0$ we focus on minimizing J_2

Weighted sum minimization as regular least squares

- We can transform the multi-objective least squares into **standard form**:

$$\min \lambda_1 ||A_1 x - b_1||^2 + \lambda_2 \, ||A_2 x - b_2||^2 + \cdots + \lambda_k \, ||A_k x - b_k||^2 \iff$$

Do the algebra as exercise!

$$\iff \quad \min \left\| \begin{pmatrix} \sqrt{\lambda_1}||A_1 x - b_1|| \\ \sqrt{\lambda_2}||A_2 x - b_1|| \\ \vdots \\ \sqrt{\lambda_k}||A_k x - b_1|| \end{pmatrix} \right\|^2 \quad \iff \quad \min \left\| \tilde{A} x - \tilde{b} \right\|^2$$

where : $\tilde{A} = \begin{bmatrix} \sqrt{\lambda_1}\, A_1 \\ \sqrt{\lambda_2}\, A_2 \\ \vdots \\ \sqrt{\lambda_k}\, A_k \end{bmatrix}$ and $\tilde{b} = \begin{pmatrix} \sqrt{\lambda_1}\, b_1 \\ \sqrt{\lambda_2}\, b_2 \\ \vdots \\ \sqrt{\lambda_k}\, b_k \end{pmatrix}$

Multi-objective least squares – Solution

$$\min \left|\left|\tilde{A}x - \tilde{b}\right|\right|^2$$

- Using the standard solution process and assuming that $\tilde{A}$ has **linear independent columns** we can find the optimal solution

$$\tilde{A}^T \tilde{A}\ x^* = \tilde{A}^T\ \tilde{b} \quad \Leftrightarrow$$

$$\Leftrightarrow \left(\lambda_1 A_1^T A_1 + \lambda_2 A_2^T A_2 + \cdots + \lambda_k A_k^T A_k\right) x^* = \left(\lambda_1 A_1^T b_1 + \lambda_2 A_2^T b_2 + \cdots + \lambda_k A_k^T b_k\right)$$

- **Remark 1**: we can use Cholesky factorization on $\tilde{A}^T \tilde{A}$

- **Remark 2**: The A_i matrices can have linearly dependent columns but $\tilde{A}$ cannot!

The optimal trade off curve – Pereto front

- Consider the **bi-objective** optimization problem

$$\min J_1(x) + \lambda J_2(x) \quad \rightarrow \quad x^*(\lambda) = \arg\min J_1(x) + \lambda J_2(x)$$

- **Remark**: The optimal solution of the bi-objective problem is a function of the parameter λ

- **Definition**: $x^*(\lambda)$ is called **Pareto optimal** iff there is no point $z \in R^n$ that simultaneously satisfies

$$J_1(z) \leq J_1\left(x^*(\lambda)\right) \quad and \quad J_2(z) \leq J_2\left(x^*(\lambda)\right)$$

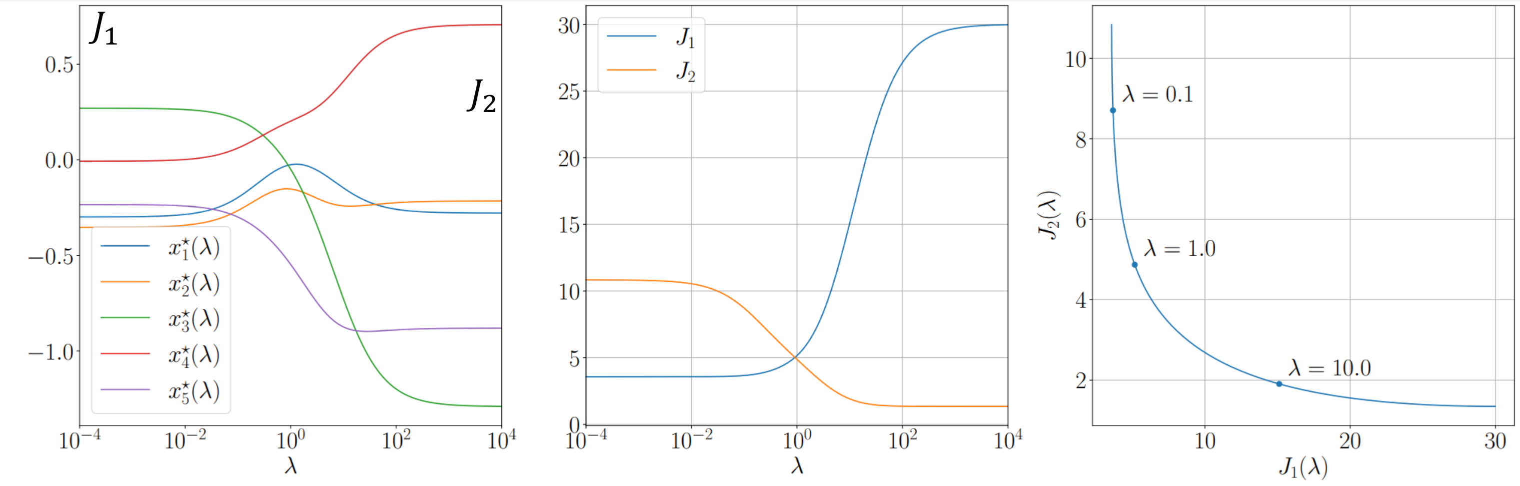
and at least one of these inequalities is strict.

- The set of points $P = \left\{\left(J_1\left(x^*(\lambda)\right), J_2\left(x^*(\lambda)\right)\right) : \ \lambda > 0\right\}$ is called the Pareto front or the Optimal trade-off curve

The optimal trade off curve – Pereto front

- **Remark**: Each point on the Pareto front represents a solution where no objective can be improved without <u>degrading</u> at least one other objective.

- **Example**: $\min J_1(x) + \lambda J_2(x)$, where $A_1, A_2 \in R^{10\times 5}$ and $x^*(\lambda) \in R^4$



NOTE: if $\boldsymbol{\lambda \approx 0}$ we essentially solve $\boldsymbol{min\,J_1(x)}$ and if $\boldsymbol{\lambda \gg 0}$ we essentially solve $\boldsymbol{min\,J_2(x)}$

Using multi-objective least squares

- **Practical guidelines for using multi-objective LS**

1. Identify the **primary objective**
   - This is your **main focus** (to minimize)
2. Choose **one or more secondary objectives**
   - These are quantities you would **prefer** to be small (e.g., size of x)
3. **Fine-tune** the weights λ's until you are **satisfied** with the solution $x^*(\lambda)$
   - **Example**: For bi-objective problems: $\min J_1(x) + \lambda J_2(x)$ $\Rightarrow$

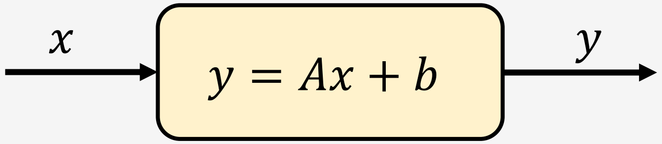
If J_2 is too big, increase λ

If J_1 is too big, decrease λ

- **An application in Control**

Control

- The **basic problem** in control theory:



- x is called the **input** or the **state** vector
- y is called the **output** or the **results** vector
- A is the **System** or **Dynamics** matrix
- b is the **offset** or **bias** vector (if $x = 0$ then $y = b$)
- By **controlling the input**, you can **achieve a desired output**

Control

- The **basic problem** in control theory:

$$x \longrightarrow \boxed{y = Ax + b} \longrightarrow y$$

- x is called the **input** or the **state** vector
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- By **controlling the input**, you can achieve a **desired output**

Example: **Coffee Roaster's Temperature Control**

State vector $x \in R^2$

- **Power setting of heaters 1 and 2**

Output vector $y \in R^2$

- **Temperature at sensors 1 and 2**

System matrix $A \in R^{2\times 2}$

- A_{ij} **how much 1-unit increase in power for heater** j **affects the temperature at sensor** i

Offset vector $b \in R^2$

- **The room temperature**, $b_1 = b_2 = 72^0F$

Goal: Choose x (which determines y) to optimize multiple objectives

Multi-objective control

- **Goal**: Choose x (which determines y) to optimize multiple objectives

$$\min J_1(x) + \lambda J_2(x)$$

- **Primary objective**:

$$J_1(x) = \left|\left|y - \boldsymbol{y^{des}}\right|\right|^2 = \left|\left|Ax + b - \boldsymbol{y^{des}}\right|\right|^2$$

where $\boldsymbol{y^{des}}$ **:** desired output

- Some common **secondary objectives**:
  - **Make x small**: $J_2(x) = \left|\left|x\right|\right|^2$
  - **Keep x close to a nominal (or safe) input**: $J_2(x) = \left|\left|x - \boldsymbol{x^{nom}}\right|\right|^2$

where $\boldsymbol{x^{nom}}$ **:** desired output

Product demand shaping

Consider n products

- $p \in R^n$ is the product **price** vector
- $d \in R^n$ is the product **demand** vector
- $\delta^{pri} \in R^n$ **fractional change in price**
$$\delta_i^{pri} = \frac{p_i^{new} - p_i}{p_i}$$
- $\delta^{dem} \in R^n$ **fractional change in demand**
$$\delta_i^{dem} = \frac{d_i^{new} - d_i}{d_i}$$
- $\boldsymbol{E^d} \in R^{n \times n}$ **demand elasticity matrix**
- Linear **demand elasticity**: $\delta^{dem} = E^d \delta^{pri}$

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Example: Linear elasticity:

We have $n = 2$ products 1 & 2, and we know the demand elasticity matrix E^d. Then the linear demand elasticity is given by

$$\begin{pmatrix} \delta_1^{dem} \\ \delta_2^{dem} \end{pmatrix} = \begin{bmatrix} -0.4 & 0.33 \\ 0.2 & -.15 \end{bmatrix} \begin{pmatrix} \delta_1^{pri} \\ \delta_2^{pri} \end{pmatrix}$$

Analysis:

- 1% increase in the price of product 1 will cause 0.4% drop in the demand of product 1, AND 0.2% increase in the demand of product 2.
- Similarly for the price change of product 2.

Product demand shaping

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- **Input (action) vector**: δ^{price}
- **Output vector**: δ^{dem}
- **Primary Objective**: find the price changes that will **not** move the demand changes too far from δ^{tar}

$$J_1(\delta^{pri}) = \left|\left|\delta^{dem} - \delta^{tar}\right|\right|^2 = \left|\left|\boldsymbol{E^d}\delta^{pri} - \delta^{tar}\right|\right|^2$$

- **Secondary Objective**: your price changes should small

$$J_2(\delta^{pri}) = \left|\left|\delta^{pri}\right|\right|^2$$

- Least squares optimization problem:

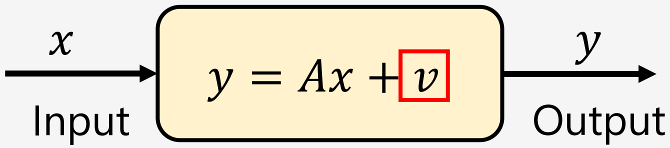
$$\mathbf{min}\, \boldsymbol{J_1}(\boldsymbol{\delta^{price}}) + \boldsymbol{\lambda J_2}(\boldsymbol{\delta^{price}})$$

- **Estimation and Inversion**

Estimation

Basic estimation problem:

We know the output y of the system and want to estimate the input x



- $y \in R^m$: **known** vector of measurements
- $A \in R^{m \times n}$: **known** matrix
- $x \in R^n$: **unknown** vector of parameters
- $v \in R^m$: **unknown** vector of noise
- **Goal**: find x given y, A, prior knowledge of x

- **Inversion**:
  - if $v = 0$ and A has linear independent columns
    $$x = A^+ y$$
- **Estimation**:
  - If $v \neq 0$, then $y = Ax + v$ does not have a unique solution.
  - Find an x that minimize the error (noise), using least squares
    $$\min J_1(x) = ||y - Ax||^2$$

Regularized estimation

x (Input) $\longrightarrow$ $y = Ax + \boxed{v}$ $\longrightarrow$ y (Output)

- **Primary objective**: $J_1(x) = ||y - Ax||^2$

- Ways to incorporate **prior information** about the **input** x, as **secondary objectives**

  - $J_2(x) = ||x||^2$ → if x should be small (aka, Tikhonov regularization)

  - $J_2(x) = ||x - x^{nom}||^2$ → if x should be close to vector x^{nom}

  - $J_2(x) = ||Dx||^2$: if x should be smooth (i.e., x_{i+1} should be close to x_i)

$$D = \begin{bmatrix} -1 & 1 & 0 & \dots & 0 & 0 & 0 \\ 0 & -1 & 1 & \dots & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & \dots & -1 & 1 & 0 \\ 0 & 0 & 0 & \dots & 0 & -1 & 1 \end{bmatrix}$$

Tikhonov regularized estimation

- Using as secondary objective $J_2(x) = ||x||^2$ we get the **bi-objective least squares**

$$\min f(x) = J_1(x) + \lambda J_2(x) = ||y - Ax||^2 + \lambda ||x||^2, \qquad where \;\; \lambda > 0$$

- The two objectives can be written as the **standard least squares problem**

$$\min f(x) = ||y - \tilde{A}x||^2, \qquad where \;\; \tilde{A} = \begin{bmatrix} A \\ \sqrt{\lambda}\, I \end{bmatrix}$$

- Then we can compute the optimal solution by solving the **normal equations**:

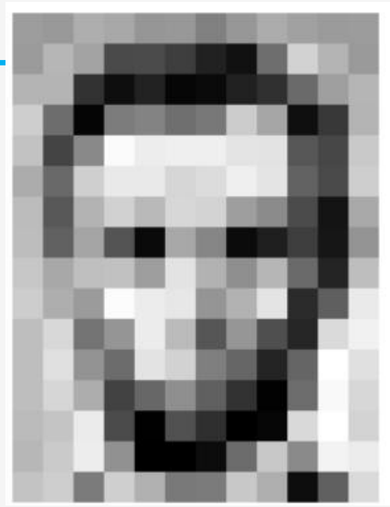
$$\tilde{A}^T \tilde{A}x = \tilde{A}^T y \;\; \Leftrightarrow \;\; (A^T A + \lambda I)x^* = A^T y$$

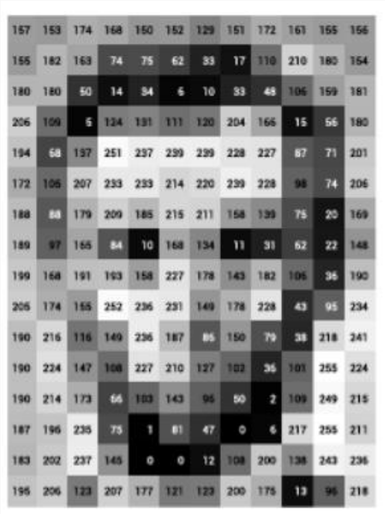
- **Remark**: the objective can be written as

$$f(x) = x^T(A^T A + \lambda I)x - 2y^T Ax + y^T y$$

- **Image representation**

Monochrome images





- **Images** can be represented as a **matrix** $X \in R^{m \times n}$
  - Each matrix element X_{ij} stores the **pixel's intensity**
  - $X_{ij} = 0$ → **black**; $X_{ij} = 1$ → **white**
    - (or $X_{ij} = 0$ → black; $X_{ij} = 255$ → white)
- An image can be **flattened** if the matrix $X \in R^{m \times n}$ is transformed into a **vector** $x \in R^{mn}$
  - **Transformation**: $X_{ij} = x_{m(j-1)+i}$ for $i = 1, \dots, m$ and $j = 1, \dots, n$

Image deblurring

- Suppose you have been given a **blurred image** $y \in R^{mn}$

$$y = Ax + u$$

  - Where $x \in R^{mn}$ is the clean image (unknown),
  - $u \in R^{m}$ is the noise vector (unknown)
  - $A \in R^{m \times n}$ is the blurring matrix (known)

- We can **recover the image** x by solving

$$\min ||Ax - y||^2 + \lambda \left(||D_v x||^2 + ||D_h x||^2 \right)$$

- The **regularization terms** are defined as

$$||D_v x||^2 = \sum_{i=1}^{m} \sum_{j=1}^{n-1} (X_{i,j+1} - X_{ij})^2$$

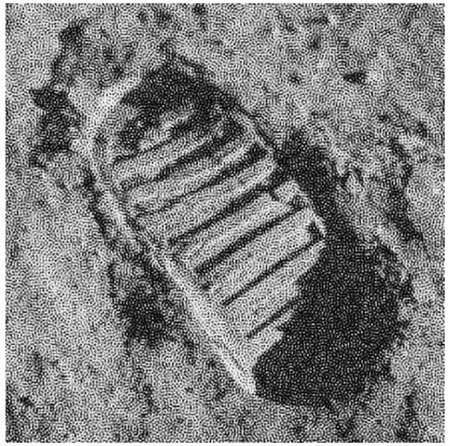
$$||D_v x||^2 = \sum_{i=1}^{m-1} \sum_{j=1}^{n} (X_{i+1,j} - X_{ij})^2$$

  - Always remember: $X_{ij} = x_{m(j-1)+i}$

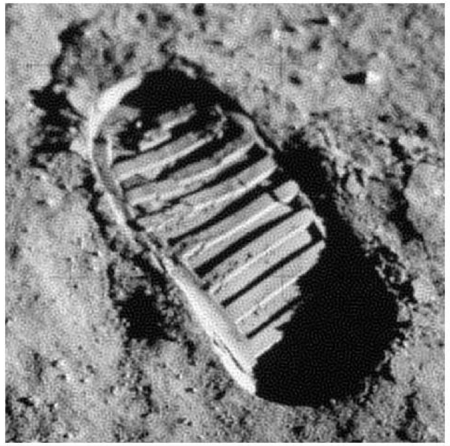
- The term: $||D_v x||^2 + ||D_h x||^2$ is called **Dirichlet energy**

Blurred image

Regularization Path



$\lambda = 10^{-6}$



$\lambda = 10^{-4}$

$\lambda = 10^{-2}$

$\lambda = 1$

[VMLS book, Page 332]

- **Regularized data fitting**

Motivation for regularization

- Recall the **data fitting model**

$$\hat{f}(x) = \theta_1 f_1(x) + \theta_2 f_2(x) + \cdots + \theta_p f_p(x)$$

  - Where $f_i(x)$ are the basis functions and θ_i are the weights/parameters

- **Intuitively**: θ_i can be thought of as the **sensitivity** of $\hat{f}(x)$ to $f_i(x)$

- Hence, we would better have $\theta_2, \theta_3, \ldots \theta_p$ **small**

  - Except θ_1 because we always have $\theta_1 = 1$

Regularized data fitting

- Suppose you have the **dataset**

$$D = \{(x^{(k)}, y^{(k)}) : x^{(k)} \in R^n, y^{(k)} \in R\,, k = 1, \dots, N\}$$

- The **training error** is: $r = A\theta - y$

- **Regularized data fitting**:

$$\min ||A\theta - y||^2 + \lambda\, ||\theta_{2:p}||^2$$

- **Ridge regression**:

$$\hat{y}^{(i)} = \left(x^{(i)}\right)^T \beta + v \quad \rightarrow \quad \hat{y} = X^T\beta + v\mathbf{1}$$

$$\min ||X^T\beta + v\mathbf{1} - y||^2 + \lambda ||\beta||^2$$

Choose λ using the **validation** process

Example

- Consider a **physical phenomenon** described by the **model**:

$$f(x) = c + \sum_{k=1}^{4} \alpha_k \sin(\omega_k x + \phi_k)$$

  - Where: x is the time, and $c = 1.54, \alpha_1 = 0.66, \alpha_2 = -0.90, \alpha_3 = 3.55, \alpha_4 = 0.89.$

  - Also, the parameters ω_k, ϕ_k are <u>known</u>, but omitted here...

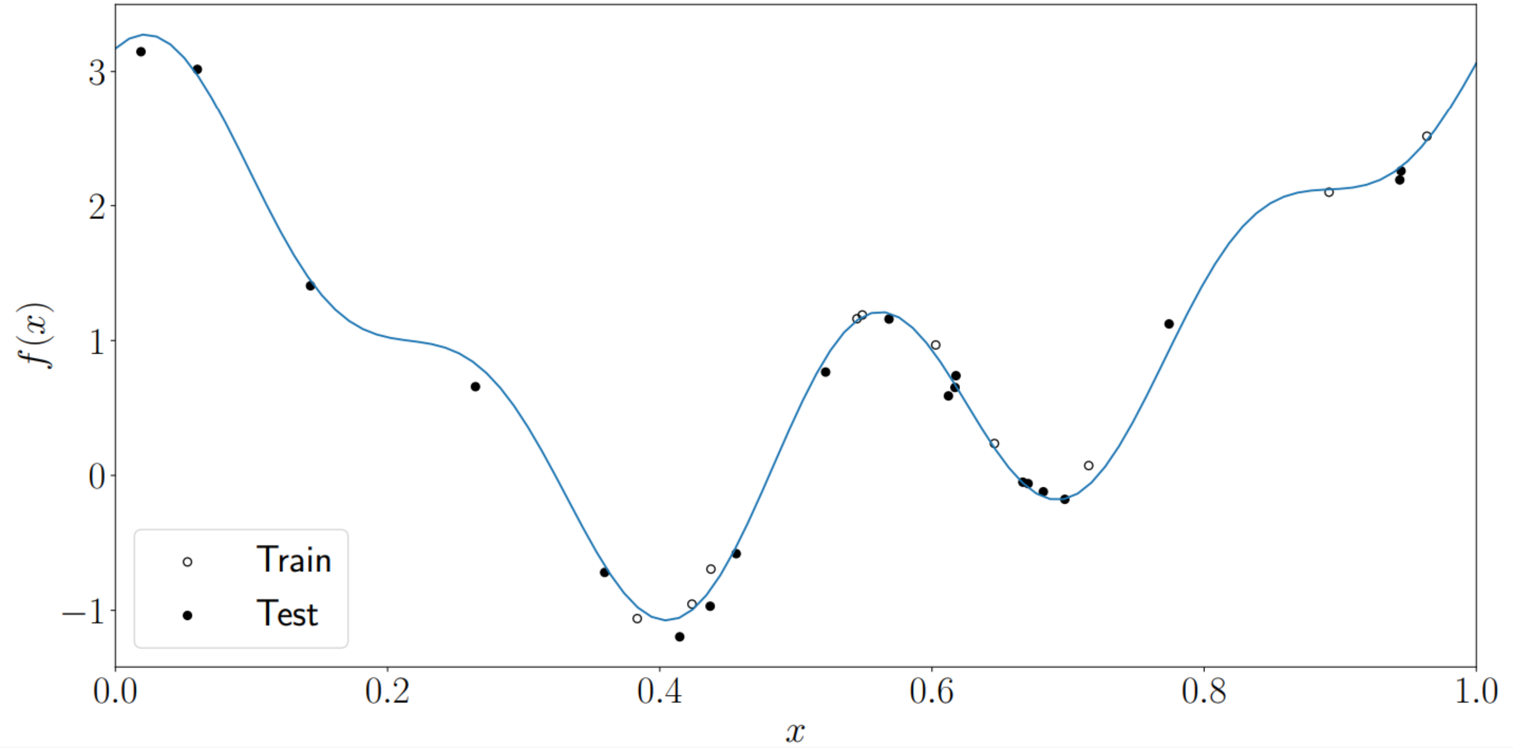
- Model $f(x)$ is **totally unknown to us**! We are only given $\boldsymbol{N} = \mathbf{10}$ **<u>noisy</u> data points** generated by $f(x)$.

- We will **fit** a model to the data of the form:

$$\hat{f}(x) = \theta_1 \boldsymbol{f_1}(\boldsymbol{x}) + \theta_2 \boldsymbol{f_2}(\boldsymbol{x}) + \theta_3 \boldsymbol{f_3}(\boldsymbol{x}) + \theta_4 \boldsymbol{f_4}(\boldsymbol{x}), \qquad where\ \boldsymbol{f_1}(\boldsymbol{x}) = \mathbf{1}, \qquad \boldsymbol{f_k}(\boldsymbol{x}) = \mathbf{sin}(\boldsymbol{\omega_k x} + \boldsymbol{\phi_k})\,, k = 2,3,4$$

- **Remark**: if $\theta_1 = c$ and $\theta_i = \alpha_i\,, i = 2,3,4$ then $f(x) = \hat{f}(x)$.

Example



The unknown model $f(x) = c + \sum_{k=1}^{4} \alpha_k \sin(\omega_k x + \phi_k)$ and the N generated noisy data

Example

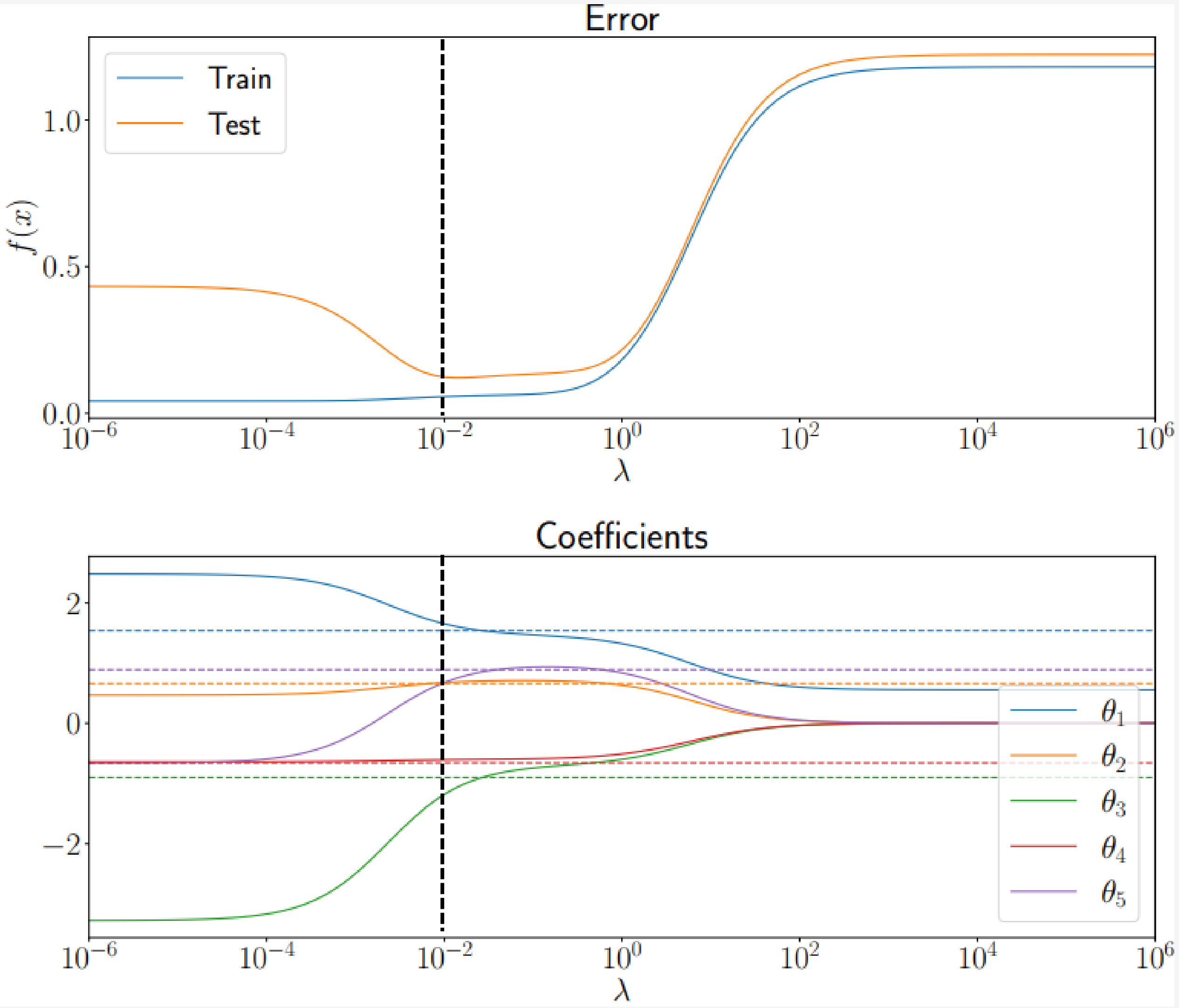
We compute the parameters $\theta_1, \theta_2 \theta_3, \theta_4$ by solving the regularized fitting problem

$$\min ||A\theta - y||^2 + \lambda ||\theta_{2:4}||^2$$

Where $A = \begin{bmatrix} 1 & f_2(x^{(1)}) & f_3(x^{(1)}) & f_4(x^{(1)}) \\ 1 & f_2(x^{(2)}) & f_3(x^{(2)}) & f_4(x^{(2)}) \\ \vdots & \vdots & \vdots & \vdots \\ 1 & f_2(x^{(N)}) & f_3(x^{(N)}) & f_4(x^{(N)}) \end{bmatrix}$, $\theta = \begin{pmatrix} \theta_1 \\ \theta_2 \\ \theta_3 \\ \theta_4 \end{pmatrix}$, $y = \begin{pmatrix} y^{(1)} \\ y^{(2)} \\ \vdots \\ y^{(N)} \end{pmatrix}$, and $\lambda > 0$

Example

- **Best model** when $\lambda \approx 0.013$ (the black dashed vertical line)
- The **dashed colored lines** represent the parameters $\alpha_1, \alpha_2, \alpha_3, \alpha_4$ of the original (unknown) model $f(x)$.
- The **solid colored lines** show the evolution of the θ's as λ increases.
- For $\lambda \approx 0.0013$ the θ's are very close to the α's
- As $\lambda \to \infty$ we have $\theta \to 0$, because the regularization term $\lambda||\theta||^2$ becomes dominant in the objective of the fitting problem



References

- S. Boyd, L. Vandenberghe: Introduction to Applied Linear Algebra – Vectors, Matrices, and Least Squares:
  - Chapter 15: Multi-objective least squares

Next lecture

- Constrained least squares